Now, $\cos\theta = \frac{1}{8}$ implies $2\cos^2\frac{\theta}{2} - 1 = \frac{1}{8}$. Thus, $\cos\left(\frac{\theta}{2}\right) = \frac{3}{4}$, since $\cos\left(\frac{\theta}{2}\right)$ is positive.

Thus, $\cos\left[\frac{1}{2}\cos^{-1}\left(\frac{1}{8}\right)\right] = \cos\left(\frac{\theta}{2}\right) = \frac{3}{4}$.

(iii) $\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$

Let $a = \tan\theta$.

Now,

$$\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right] = \tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)\right]$$

$$= \tan\left[\frac{1}{2}\sin^{-1}(\sin 2\theta) + \frac{1}{2}\cos^{-1}(\cos 2\theta)\right] = \tan[2\theta] = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2a}{1-a^2}.$$

Example 4.19

Prove that $\tan(\sin^{-1}x) = \frac{x}{\sqrt{1-x^2}}$ for $|x| < 1$.

Solution

Let $\sin^{-1}x = \theta$. Then, $x = \sin\theta$ and $-1 \le x \le 1$

Now, $\tan(\sin^{-1}x) = \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{\sqrt{1-\sin^2\theta}} = \frac{x}{\sqrt{1-x^2}}$, $|x| < 1$.

Example 4.20

Evaluate $\sin\left[\sin^{-1}\left(\frac{3}{5}\right) + \sec^{-1}\left(\frac{5}{4}\right)\right]$

Solution

Let $\sec^{-1}\frac{5}{4} = \theta$. Then, $\sec\theta = \frac{5}{4}$ and hence, $\cos\theta = \frac{4}{5}$.

Also, $\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\left(\frac{4}{5}\right)^2} = \frac{3}{5}$, which gives $\theta = \sin^{-1}\left(\frac{3}{5}\right)$.

Thus, $\sec^{-1}\left(\frac{5}{4}\right) = \sin^{-1}\left(\frac{3}{5}\right)$ and $\sin^{-1}\frac{3}{5} + \sec^{-1}\left(\frac{5}{4}\right) = 2\sin^{-1}\left(\frac{3}{5}\right)$.

We know that $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\sin^{-1}x$, if $|x| \le \frac{1}{\sqrt{2}}$.

Since $\frac{3}{5} < \frac{1}{\sqrt{2}}$, we have $2\sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(2\times\frac{3}{5}\sqrt{1-\left(\frac{3}{5}\right)^2}\right) = \sin^{-1}\left(\frac{24}{25}\right)$.

Hence, $\sin\left[\sin^{-1}\left(\frac{3}{5}\right) + \sec^{-1}\left(\frac{5}{4}\right)\right] = \sin\left(\sin^{-1}\left(\frac{24}{25}\right)\right) = \frac{24}{25}$, since $\frac{24}{25} \in [-1, 1]$.

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